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Airborne hyperspectral imaging of cover crops through radiative transfer process-guided machine learning

Sheng Wang[1,2*], Kaiyu Guan[1,2,3,4,5*], Chenhui Zhang[1,3,4], Chongya Jiang[1,2], Qu Zhou[1,2], Kaiyuan Li[1,2], Ziqi Qin[1,2], Elizabeth A. Ainsworth[1,2,6,7], Jingrui He[3,4,5], Jun Wu[3,4], Dan Schaefer[8], Lowell E. Gentry[2], Andrew J. Margenot[1,2], Leo Herzberger[1,2]

1. Agroecosystem Sustainability Center, Institute for Sustainability, Energy, and Environment, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

2. Department of Natural Resources and Environmental Sciences, College of Agricultural, Consumer and Environmental Sciences, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA

3. National Center for Supercomputing Applications, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

4. Department of Computer Science, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

5. School of Information Sciences, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

6. Department of Plant Biology, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

7. USDA ARS Global Change and Photosynthesis Research Unit, Urbana, IL 61801, USA

8. Illinois Fertilizer and Chemical Association, Bloomington, IL 61701, USA

*Email: shengwang12@gmail.com; kaiyug@illinois.edu

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Abstract:

Cover cropping between cash crop growing seasons is a multifunctional conservation practice. Timely and accurate monitoring of cover crop traits, notably aboveground biomass and nutrient content, is beneficial to agricultural stakeholders to improve management and understand outcomes. Currently, there is a scarcity of spatially and temporally resolved information for assessing cover crop growth. Remote sensing has a high potential to fill this need, but conventional empirical regression operated with coarse-resolution multispectral data has large uncertainties. Therefore, this study utilized airborne hyperspectral imaging techniques and developed new process-guided machine learning approaches (PGML) for cover crop monitoring. Specifically, we deployed an airborne visible to shortwave infrared (400–2400 nm) hyperspectral system to acquire high spatial (0.5 m) and spectral (3–5 nm) resolution reflectance over 23 cover crop fields across Central Illinois in March and April of 2021. Airborne hyperspectral surface reflectance with high spectral and spatial resolution can be well matched with field data to quantify cover crop traits. Furthermore, the PGML models were pre-trained by synthetic data from soil-vegetation radiative transfer modeling (1,000,000 records), and then fine-tuned with field data of cover crop biomass and nutrient content. Results show that airborne hyperspectral data with PGML can achieve high accuracy to predict cover crop aboveground biomass (R^2 = 0.72, relative RMSE = 15.16%) and nitrogen content (R^2 = 0.69, relative RMSE = 16.59%) through leave-one-field-out cross-validation. Unlike the pure data-driven approach (e.g., partial least-squares regression), PGML incorporated radiative transfer knowledge and obtained higher predictive performance with fewer field data. Meanwhile, with field data for model fine-tuning, PGML predicted biomass more accurately than the numerical inversion of radiative transfer models. We also found that the red edge has a high contribution in quantifying aboveground biomass and nitrogen content, followed by green and shortwave spectra. This study demonstrated the first attempt of utilizing hyperspectral remote sensing to accurately quantify cover crop traits. We highlight the strength of PGML in exploiting sensing data to advance agroecosystem monitoring for sustainable agricultural management.

Keywords: Cover crop, Aboveground biomass, Nitrogen, Radiative transfer modeling, Process-guided machine learning, Imaging spectroscopy

1. Introduction

Cover crops grown in the winter fallow period between cash crop rotations are essential for agroecosystem conservation. Cover cropping can bring significant benefits of reducing soil erosion and nutrient loss (Hively et al., 2020; Sharma et al., 2018), improving soil health (Wood and Bowman, 2021), facilitating carbon sequestration (Qin et al., 2021), and promoting cropland biodiversity (Elhakeem et al., 2019). These benefits are vital for improving agroecosystem sustainability, particularly for the U.S. Corn Belt croplands that contribute one-third of global corn and soybean production (USDA, 2020) but suffer from soil degradation (Thaler et al., 2021). As such, federal and state conservation programs have been launched to provide financial and technical support to promote cover crop adoption in the U.S. (Thieme et al., 2020), and cover cropping acreage has increased by 50% from 2012 to 2017 (Wallander et al., 2021). Beyond cover crop adoption, the accurate spatially and temporally resolved information of cover crop growth traits, specifically aboveground biomass and nutrient status, is also highly important for assessing the outcomes of these conservation programs. However, such high-resolution information for cover crop growth is largely missing, and effective approaches to accurately quantify cover crop growth are urgently needed.

Besides assessing outcomes of conservation programs, the accurate information of cover crop growth, such as aboveground dry biomass, plant nitrogen concentration, nitrogen content, and carbon to nitrogen ratio (C/N), is beneficial to growers for cropland management. Cover cropping brings both pros and cons to the existing row crop systems, and cover crop growth information is highly needed to guide growers for termination (Plastina et al., 2020; Vincent-Caboud et al., 2017). Planted after cash crop harvesting, cover crops can uptake unutilized soil inorganic nitrogen and thus reduce leaching losses, and can also

77 protect soils from water or wind erosion (Lee and McCann, 2019). However, early termination or late
78 planting of cover crops leads to insufficient cover crop growth and reduced benefits for agroecosystem
79 conservation. For example, to reduce 80% soil nitrate, cover crops need to have biomass > 100 $g \cdot m^{-2}$
80 (Hively et al., 2009). Meanwhile, cover crops need to be terminated in time to avoid over-consumption of
81 moisture and nutrients in the soil, which could compromise the following cash crop growth (Noland et al.,
82 2018). Cover crop termination time affects subsequent tissue decomposition and soil nitrogen
83 mineralization for supporting the coming cash crop growth. As such, too late or too early termination of
84 cover crops could either threaten the coming cash crop yield potential or reduce agroecosystem
85 conservation benefits, respectively (Qin et al., 2021). Timely information on cover crop biomass and
86 nutrients is urgently needed to inform growers of the appropriate time to terminate cover crops (Fan et al.,
87 2020). Furthermore, cover crop biomass and nutrient information can be integrated into agroecosystem
88 models (e.g., Qin et al., 2021) to quantify cropland nutrient and carbon dynamics as well as quantify
89 cropland carbon credits.
90
91 As cover crop growth is highly dynamic in space and time, deriving high-resolution accurate cover crop
92 traits faces challenges because of lacking cost-effective approaches. Conventional field destructive
93 sampling and laboratory analysis to quantify cover crop traits of biomass and nutrient content are labor-
94 intensive and insufficient to achieve large spatial and temporal coverage. Remote sensing provides a
95 promising opportunity to quantitatively assess cover crop traits by surface reflectance. For example,
96 studies have demonstrated the potential of multispectral data from satellites (Breunig et al., 2020; Fan et
97 al., 2020; Hively et al., 2009; Thieme et al., 2020) or airborne systems (Yuan et al., 2019) to derive
98 vegetation indices such as Normalized Difference Vegetation Index (NDVI) to quantify cover crop
99 aboveground biomass (Muñoz et al., 2010; Prabhakara et al., 2015) and nitrogen content (Hively et al.,
100 2009; Yuan et al., 2019). In single- or multi-site studies (<10 fields), high accuracy R^2 of up to 80% could
101 be achieved to predict cover crop biomass or nitrogen content (Biewer et al., 2009; Prabhakara et al.,
102 2015; Xu et al., 2018; Yuan et al., 2019). However, studies across more than 10 sites are rare and usually

have limited accuracy, e.g., R^2 50–73%, on utilizing multispectral sensing for quantifying cover crop biomass (Goffart et al., 2021; Hively et al., 2009; Xia et al., 2021). Such accuracies are partially due to the limitations of multispectral sensing, in particular: (i) limited spectral signatures to detect plant nutrient traits, (ii) scale mismatch with field data from small sampling plots, and (iii) empirical regression lacking scalability and transferability across real-world diversity of field conditions.

Beyond multispectral sensing, the rapid progress in hyperspectral remote sensing enabled the more accurate retrieval of biophysical and biochemical plant traits (Singh et al., 2015; Zarco-Tejada et al., 2021), but studies of using hyperspectral sensing for cover crop monitoring are limited. A few pilot evaluations of ground portable sensors to collect hyperspectral reflectance in visible and near-infrared ranges have demonstrated the high accuracy of hyperspectral data to quantify cover crop traits (Hively et al., 2009; Zhao et al., 2010; Suzuki et al., 2010). Airborne hyperspectral sensing can offer surface reflectance with hundreds of spectral wavelengths and high spatial resolution (~sub-meter). Such high-value data provide a high potential to mitigate the three challenges in conventional multispectral sensing for tracking cover crop nutrient and carbon variabilities in the landscape scale (coverage < 100 km^2). Specifically, the full-optical range airborne hyperspectral reflectance (400–2400 nm) contains rich spectral information, which is closely linked with crop biochemical and biophysical traits (Wang et al., 2021a). Such high-dimensional spectral data can potentially provide accurate estimates of cover crop traits as well as a comprehensive analysis of the linkage between spectral signatures and cover crop traits. Furthermore, the high-resolution airborne imagery down to the sub-meter level can match well with cover crop field measurements from small sampling plots (typically <1 m^2, Xia et al., 2020) to solve the spatial mismatch issue. Regarding empirical regression lacking scalability and transferability in conventional multispectral sensing (Muñoz et al., 2010; Xia et al., 2020), airborne hyperspectral sensing provides high-value data sources to test new algorithms for the retrieval of plant traits from remote sensing.

As an emerging paradigm, Process-Guided Machine Learning (PGML) incorporates physical principles and domain knowledge into data-driven models and shows high capability for solving ill-posed inverse problems of differential equation models (Karniadakis et al., 2021), but has been rarely tested for the retrieval of plant traits. Conventional sensing-based retrievals of plant traits can be categorized into four groups: parametric regression, non-parametric regression, physically-based model inversion, and hybrid regression (Verrelst et al., 2019). With hundreds of spectral wavelengths, hyperspectral data are often operated with data-driven non-parametric regression (e.g., partial least-squares regression, PLSR) and soil-vegetation radiative transfer models (RTM) with numerical inversion (e.g., van der Tol et al., 2016) or machine learning emulators (e.g., Verrelst et al., 2016; Wang et al., 2021b). Particularly, machine learning emulators or hybrid machine learning approaches, which pretrain the decision tree models (Wang et al., 2021b), neural networks (Danner et al., 2021), or kernel-based regression models (Berger et al., 2020a) with RTM simulations, can be applied efficiently for large-volume remote sensing observations. However, hybrid approaches still face challenges including data noise in remote sensing observations, RTM model imperfection, and poor performance in training machine learning over large randomly sampled RTM synthetic data (Danner et al., 2021; Verrelst et al., 2021). To mitigate these challenges, recent studies focused on intelligent feature space sampling (Verrelst et al., 2021), prior knowledge to reduce RTM parameters (Wang et al., 2021b), transfer learning (Wu et al., 2022; Zhang et al., 2021), and dimensionality reduction to mitigate data noises (Danner et al., 2021). As an emerging machine learning paradigm to fully integrate data-driven and process-based approaches, PGML provides a potential solution to enable remote sensing retrieval to be more transferable and scalable across diverse conditions.

The low aboveground biomass due to the winter dormancy of cover crops in the U.S. Corn Belt (Barnes et al., 2021) poses a challenge to remote sensing quantification of traits. Additionally, remotely detected signals from cover crop fields are often mixed with various background signals from soils and crop residues (Hively et al., 2009; Thieme et al., 2020). Traditional data-driven approaches such as PLSR are usually operated with pure vegetation signals (e.g., NDVI > 0.65, Wang et al., 2021b) and lack robustness

for retrieving plant traits in partially vegetated conditions as soil backgrounds may vary significantly across and within fields. To achieve accurate retrieval of crop traits, data-driven approaches require a sufficient amount of representative data for model training. However, field data are often difficult to acquire. In contrast, RTM retrieval could be more robust, generalized, and transferable to be applied across soil conditions and plant growth stages as RTMs are developed based on photon transport laws and radiative transfer theories. RTM retrieval can directly operate with sensing data without measured plant traits as labels. However, RTMs often have weaker predictive performance than the data-driven approaches (Wang et al., 2021a, b) as the incomplete knowledge of the process (Verhoef and Bach, 2007) and model simplification for reducing computational demand (Jacquemoud et al., 2009). Furthermore, as defined by ordinary differential equations, RTMs have quite a number of parameters, and the equifinality of these parameters can often lead to ill-posed model inversion (Wang et al., 2021b). Therefore, the integration of process-based and data-driven approaches through PGML, which can potentially reduce the demand for field data as well as maintain a high accuracy to quantify cover crop traits, is highly needed and needs exploration.

The general objective of this study was to demonstrate the capability of airborne hyperspectral imaging of cover crop growth, and to develop PGML based on soil-vegetation radiative transfer modeling for the retrieval of cover crop traits in typical corn and soybean fields of the U.S. Corn Belt. The specific objectives were to (1) develop PGML algorithms and evaluate their accuracy for quantifying cover crop aboveground biomass and nitrogen content from airborne hyperspectral remote sensing imagery; (2) evaluate the benefits of PGML to quantify cover crop traits compared to conventional pure data-driven machine learning (i.e. PLSR) and process-based soil-vegetation radiative transfer modeling; (3) understand the key spectral signatures that contribute to remote sensing quantification of cover crop traits and assess the spectral linkage among cover crop traits.

2. Data and Methods

2.1 Study region

We conducted this study over cover crop fields in central Illinois, which include counties of Champaign, Douglas, Moultrie, Piatt, De Witt, McLean, and Ford (Fig. 1). Located in the heart of the U.S. Corn Belt (Fig. 1a), more than 90% of central Illinois is covered by corn-soybean rotation fields. Growing cover crops is an important management practice for this region to improve agricultural sustainability and resilience. The cover crop types in this region are mostly cereal rye (*Secale cereale* L., Fig. 1b, c, d), which is well suited to overwintering in the cold off-season months (Dozier et al., 2017). In fields of this study, cover crops were typically planted in September–November 2020 and terminated in April–June 2021. This region has a continental climate with mean winter temperatures of -8 °C and winter precipitation of 350 mm. The major soil order is Mollisols, with surface soil organic carbon concentration of 1–4% and soil texture of silt loam.

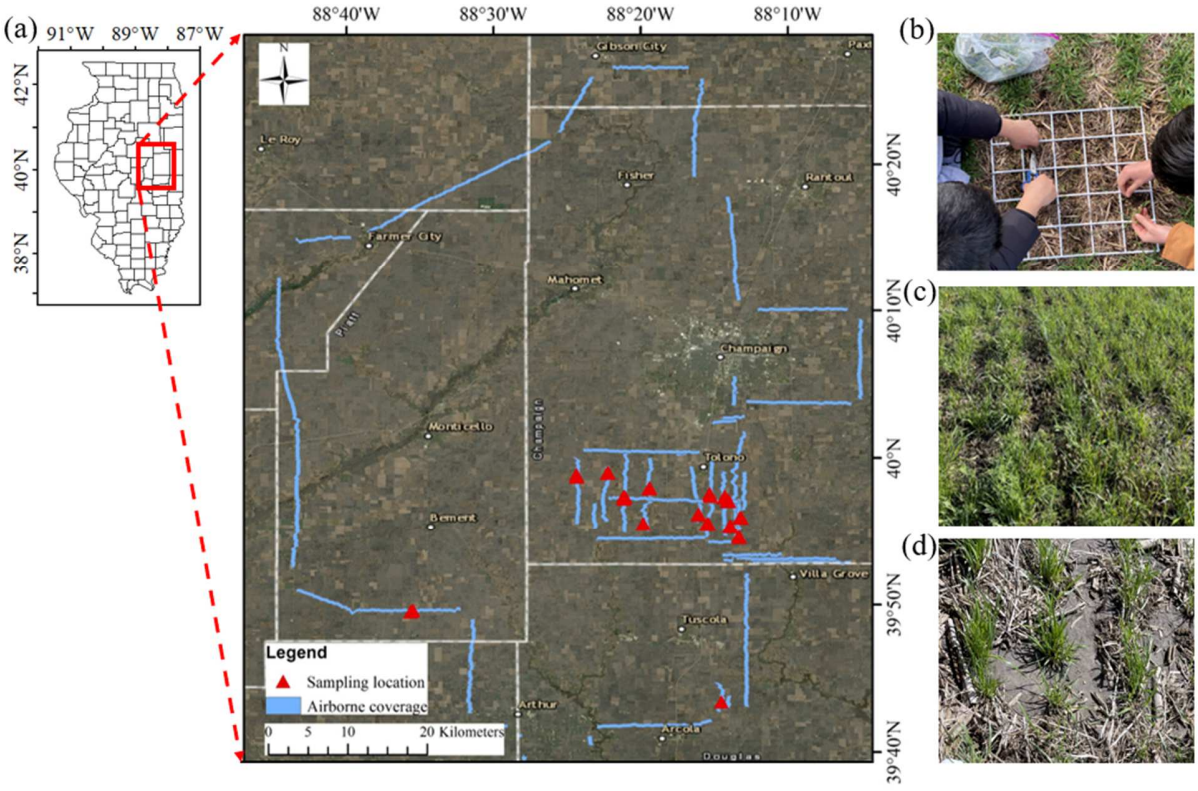


Fig. 1 (a) Study region, airborne collected hyperspectral imagery, and field measurement locations in central Illinois. The red triangles refer to the field sampling locations. The blue polygons are the covered areas by airborne flight campaigns. The base map is the Sentinel-2 imagery over the study region acquired on March 31st, 2021. (b) sampling quadrat for destructive sampling in fields. (c) field with high cover crop aboveground biomass. (d) field with low cover crop aboveground biomass.

2.2 Field measurements

Cover crop aboveground plant tissues and biomass were collected from 23 fields across the study region (Fig. 1) within three days following airborne flight campaigns (Table 1) in March and April of 2021. Fields enrolled for this study were cover crop experimental trials and private commercial fields, which have a good representation of crop growth conditions and soil properties. All of these 23 sampled fields were planted with cereal rye (*Secale cereale* L.), which is one of the most abundant cover crop species in the Corn Belt. For cover crop sampling, we randomly selected 3–4 sampling units (0.5m by 0.5m quadrat, Fig. 1b) in each cover crop field. In each quadrat, we destructively collected all aboveground cover crop tissues. Cover crop tissues were dried for two weeks at 60 °C, and we used an analytical balance (ME204TE/00, Mettler Toledo, US) to weight the total aboveground dry biomass (BM, $g \cdot m^{-2}$). Furthermore, the dried cover crop tissues were ground (Geno Grinder, BT&C Lebanon, US) into powders to analyze plant tissue nitrogen concentration through a dry-combustion elemental analyzer (Costech 4010, Costech Analytical Technologies, US). The plant nitrogen content was determined by the product of aboveground dry biomass and plant nitrogen concentration. Following the terminology of previous studies (Berger et al., 2020b; Wang et al., 2021b), the plant mass-based measure (plant nitrogen mass per dry mass) was denoted as nitrogen concentration (N_{mass}, %). The plant-level area-based measure (plant nitrogen content per ground area or nitrogen uptake, i.e. BM × N_{mass}) was denoted as canopy nitrogen content (BMn, $g \cdot m^{-2}$). Similarly, from the elemental analyzer, we also obtained cover crop plant tissue

carbon concentration (C_{mass}, %), and further used the product of Biomass and C_{mass} to derive the canopy carbon content (BMc, $g \cdot m^{-2}$). The details of these measured cover crop traits are summarized in Table 1. In this study, we have totally collected 73 cover crop biomass measurements and 63 cover crop nitrogen and carbon concentration measurements from fields with diverse cover crop growth conditions, crop residue cover and soil types. Fig. 1 c and d show examples of cover crop fields with high and low aboveground biomass, respectively. To understand the variabilities of cover crop traits, we first calculated the coefficient of variations (CV, the ratio between the standard deviations and means of measured variables) for BM, C_{mass}, BMc, N_{mass}, BMn, and C/N. To explore the relationship among these cover crop traits, we also conducted a correlation analysis and a principal component analysis (PCA). Specifically, we calculated a correlation matrix among all cover crop traits. In the PCA analysis, we visualized the first two components to compare the similarities of cover crop traits.

Table 1. Summary of measured cover crop traits, their abbreviations, units and measurement methods in this study.

Measured cover crop traits	Abbreviation	Unit	Measurement methods
Aboveground dry biomass	BM	$g \cdot m^{-2}$	Dry weight per ground area
Aboveground biomass carbon concentration	C_{mass}	%	Dry-combustion element analyzer
Aboveground biomass carbon content	BMc	$g \cdot m^{-2}$	BM * C_{mass}
Aboveground biomass nitrogen concentration	N_{mass}	%	Dry-combustion element analyzer
Aboveground biomass nitrogen content	BMn	$g \cdot m^{-2}$	BM * N_{mass}
Carbon to nitrogen ratio	C/N	-	C_{mass}/N_{mass}

2.3 Airborne hyperspectral data

We conducted two airborne hyperspectral flight campaigns on March 20th and April 13th, 2021 under sunny weather conditions (Table 2) over the study region (Fig. 1). The airborne flights were conducted with an aircraft (PA23-160, Piper Aircraft, US) equipped with HySpex VNIR-1800 (covering 400–1000 nm) and SWIR-384 (covering 950–2400 nm) hyperspectral imaging sensors (HySpex NEO, Norway).

VNIR-1800 and SWIR-384 have spectral resolutions of 3.26 and 5.45 nm, respectively. These two sensors have the same radiometric resolution of 16 bits and the similar field of views for around 17 degrees. The flights were conducted at an altitude of 1.5 km above the ground with an imagery ground resolution of 0.5 m. The campaigns were conducted with low solar zenith angles at the local time of 10:30–14:00. In the flight, the airborne system was equipped with an Inertial Measurement Unit (IMU) and differential Global Navigation Satellite Systems (GNSS) (Applanix, Trimble, US) to record the sensors' angular and position information for further image geometric processing.

In airborne hyperspectral data processing, we used the same procedures as Wang et al. (2021b) for geometric, radiometric, and atmospheric corrections. Specifically, in the radiometric calibration, airborne raw sensor imagery was converted to at-sensor radiance using laboratory calibration coefficients. Then, we used PARGE and ATCOR software (Richter and Schläpfer, 2016) to process the radiance imagery through geometric and atmospheric corrections. To conduct the geometric correction, two imaging sensor boresight coefficients, flight GNSS and IMU data, and surface elevation model were used as inputs for PARGE to process raw airborne data to be the orthorectified imagery. Then, hyperspectral imagery were processed through ATCOR, which is based on the atmospheric radiative transfer model MODTRAN look-up table simulations (Berk et al., 2005), in the next step for SMILE and atmospheric effect corrections. In atmospheric corrections, this study utilized ATCOR with selecting the mid-latitude, winter, and rural aerosol model. With all these steps, we obtained high-quality surface reflectance of 356 spectral bands ranging from 400 to 2400 nm with 3–5 nm spectral resolution and 0.5 m spatial resolution. Regarding the details on hyperspectral data processing, please refer to Wang et al. (2021b).

Table 2. Summary of airborne hyperspectral campaigns for cover crop monitoring. RH refers to relative humidity (%). Ta is the air temperature (°C).

Date	Local time	Weather	Ta (°C)	RH (%)	Cover crop growth conditions
March 20th, 2021	10:10–14:53	Sunny	12.22	45.26	Moderate biomass

April 13th, 2021	10:21–15:23	Sunny	15.31	41.03	High biomass

2.4 Quantification of cover crop traits

To develop a transferable and scalable approach, we designed the PGML framework (Fig. 2c), which include two steps: developing a pretrained machine learning surrogate model (RTMpt) from RTM synthetic datasets (1,000,000 records), and fine tuning RTMpt with measured cover crop traits to obtain the PGML model. We further compared PGML with RTMpt and two conventional approaches (Fig. 2a and b) on using hyperspectral data to quantify cover crop traits. Specifically, these two conventional approaches included pure data-driven machine learning (i.e., partial least-squares regression, PLSR, Fig. 2a) and the radiative transfer model (RTM)-based numerical optimization (RTMn, Fig. 2b).

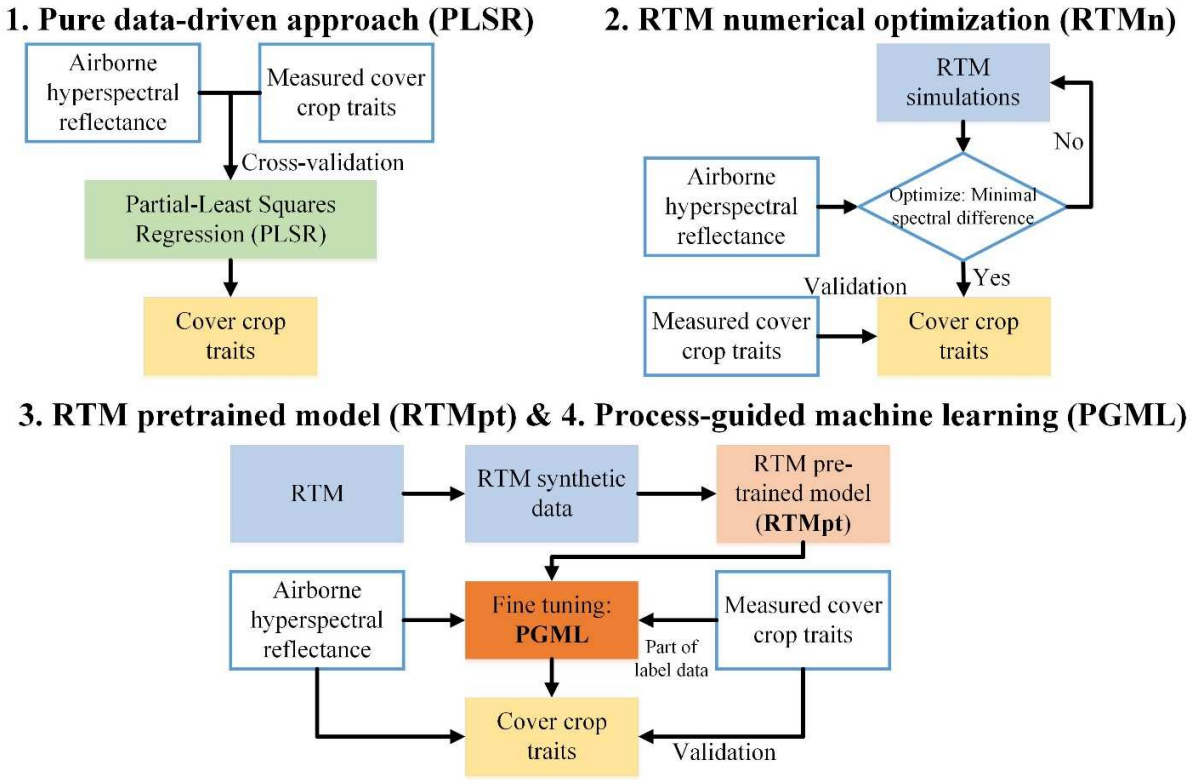


Fig. 2 Four approaches of quantifying cover crop traits from airborne hyperspectral surface reflectance. PLSR: partial least-squares regression. RTM: soil-vegetation radiative transfer modeling. RTMn: RTM retrieval through numerical optimization. RTMpt: RTM retrieval through a pre-trained machine learning

model, which used long short-term memory networks. Process-guided machine learning (PGML) is based on RTMpt with field measurements for model fine-tuning.

2.4.1 Partial least-squares regression

We first applied the data-driven approach i.e., PLSR (e.g., Ainsworth et al., 2014) to develop statistical models between airborne hyperspectral reflectance and measured cover crop traits (BM, N_{mass}, BMn, C_{mass}, BMc, and C/N ratio). PLSR is the most extensively used regression approach for remote sensing retrieval of vegetation traits (e.g., Wang et al., 2021b). Different from other statistical or machine learning approaches requiring large datasets for model development, PLSR is highly suitable for cases with small sizes of label data, as PLSR feature inputs are minimized to very few orthogonal latent components (Geladi and Kowalski, 1986; Wold et al., 2001). Furthermore, with a high degree of transparency, PLSR can provide the interpretation of how the spectral wavelengths are linked with cover crop traits. Thus, we used PLSR to develop statistical models between hyperspectral data and cover crop traits with leave-one-field-out cross-validation (e.g., Stevens et al., 2012). Data from the same field on the same date could be very similar, and if data from the same field and date were split into both training and testing, the model performance could be inflated (Azzari et al., 2019). To avoid such cases, field data were grouped into different folds according to their measurement fields and dates. In leave-one-field-out cross-validation, the selected one field data were in turn used for model testing, and other fold data were used for model training as shown in Fig. 3a. To minimize the overfitting between spectra and cover crop traits, we followed Meerdink et al. (2016) to determine the number of PLSR latent components by optimizing the prediction residual sum of squares (PRESS) statistics. Specifically, we identified the component number by minimizing PRESS with no significant differences in PLSR accuracy with additional components (e.g., Ainsworth et al., 2014). To understand the key wavelengths for predicting cover crop traits, we also calculated the Variable Influence on Projection (VIP) scores (Wold et al., 2001) from the PLSR models. Furthermore, we conducted experiments with variable training data sizes to evaluate their impacts on PLSR performance (Fig. 3b). Specifically, we used a range of training data sizes (from 10% to 90% of the

whole dataset, 10% increase in each experiment) to train PLSR models and used the rest of the dataset to test the PLSR model performance. To reduce randomness in data splitting, we conducted 20 times of random training and testing data splitting in each experiment. The comparison of the results from these experiments can provide insights into the relationships between training data sizes and PLSR model performance.

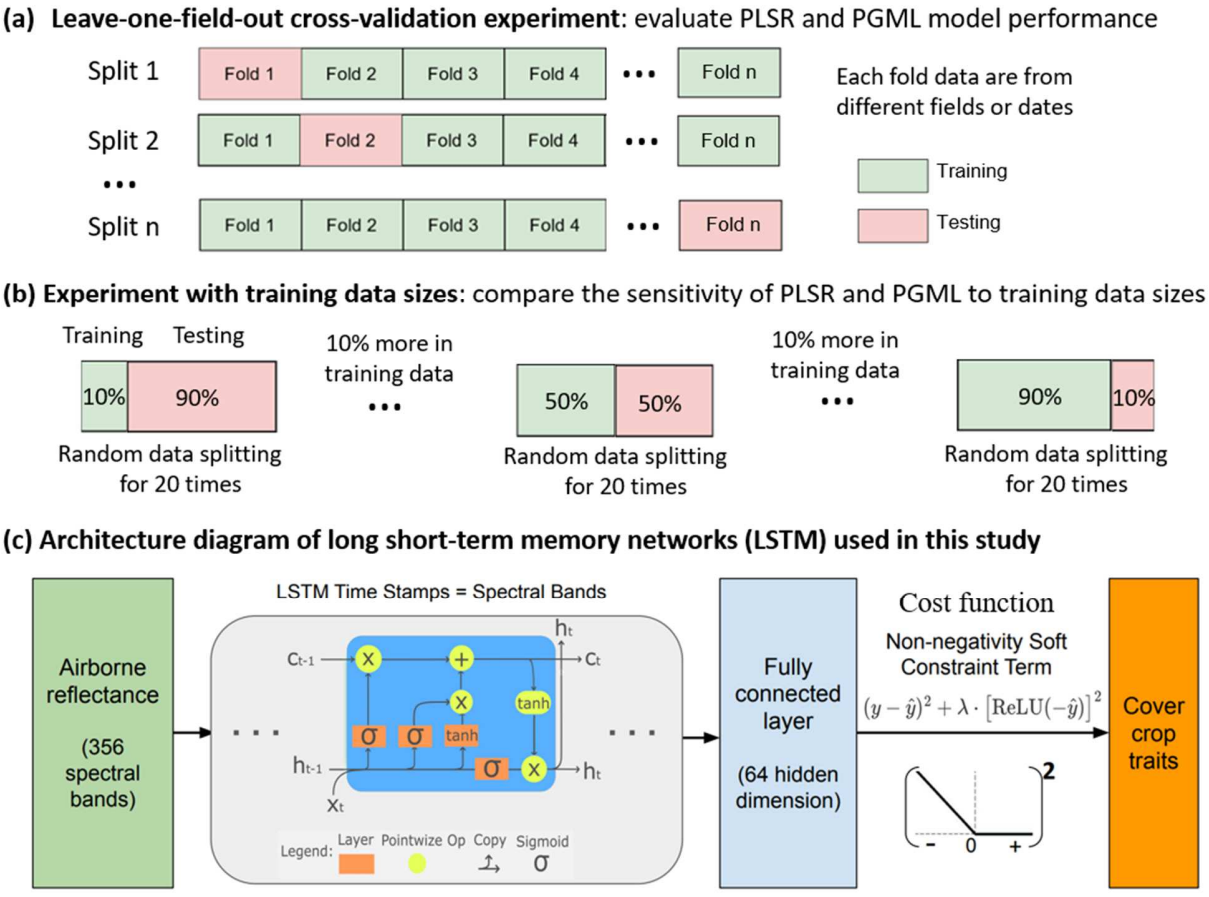


Fig. 3. Experiments to evaluate partial least-squares regression (PLSR) and process-guided machine learning (PGML) model performance and their sensitivity to training data sizes. (a) Leave-one-field-out cross-validation experiment. To avoid splitting data from the same field and date into both training and testing, field data were grouped into different folds according to their measurement fields and dates. In cross-validation, the selected one-fold data were in turn used for model testing, and other fold data were

used for model training. (b) Experiment with different training data sizes from 10% to 90% of the whole dataset. (c) Architecture diagram of long short-term memory networks (LSTM) for PGML in this study.

2.4.2 Soil-vegetation radiative transfer modeling

In soil-vegetation radiative transfer modeling, we used the optical radiative transfer module in Soil Canopy Observation Photochemistry and Energy fluxes model (SCOPE) (van der Tol et al., 2016; van der Tol et al., 2009). The optical RTM includes the leaf radiative transfer model PROSPECT-D (Féret et al., 2017; Jacquemoud et al., 2009; Jacquemoud and Baret, 1990), the canopy radiative transfer model 4SAIL (Verhoef, 1984; Verhoef and Bach, 2007), and the soil Brightness Shape Moisture model BSM (Jiang and Fang, 2019; Verhoef et al., 2018; Yang et al., 2020), and can simulate surface hyperspectral reflectance of 400–2400 nm. As shown in Table 3, PROSPECT-D used a variety of leaf traits, e.g., leaf chlorophyll content (C_{ab}), leaf dry matter content (C_{dm}), and water content, to simulate leaf reflectance, absorptance, and transmittance. 4SAIL used leaf area index (LAI), leaf angle distribution parameters (LIDFa, LIDFb), and sun-target-observer angles to simulate bidirectional reflectance on the top of the canopy. In the soil part, the global soil spectral vector model BSM can simulate reflectance signals of diverse soils. For details on parameters, please refer to Table 3. As this study primarily focused on biomass and nitrogen traits, we only utilized SCOPE to retrieve C_{ab}, C_{dm}, and LAI from the collected airborne hyperspectral reflectance.

328 **Table 3**. Parameters and their ranges in the SCOPE radiative transfer model for numerical optimization
329 and generating synthetic data to develop the pretrained machine learning model (RTMpt), which is further
330 fine-tuned with field data to develop the PGML model.

Parameter		Description	Units	Lower limit	Upper limit
Leaf submodel (PROSPECT-D)	**N**	Mesophyll structure parameter	-	1	2
	C_{ab}	Leaf chlorophyll content a and b	$\mu g \cdot cm^{-2}$	0	100
	C_{ca}	Leaf carotenoid content	$\mu g \cdot cm^{-2}$	0	25
	C_{ant}	Leaf anthocyanin content	$\mu g \cdot cm^{-2}$	0	5
	C_s	Senescent material	-	0	1.2
	C_w	Equivalent leaf water thickness	cm	0	0.24
	C_{dm}	Leaf dry matter per area	$g \cdot cm^{-2}$	0	0.02
Canopy submodel (4SAIL)	**LAI**	Canopy leaf area index	$m^2 \cdot m^{-2}$	0	8
	LIDFa	Averaged leaf slope	-	-1	1
	LIDFb	Distribution bimodality	-	-1	1
Soil submodel (BSM)	**B**	Soil brightness	-	0	0.9
	BSMlat	BSM model parameter latitude	-	10	60
	BSMlon	BSM model parameter longitude	-	10	60
	SMC	Volumetric soil moisture content	%	5	55

331

332 In the model inversion, this study utilized the numerical optimization method to retrieve cover crop traits
333 from airborne hyperspectral surface reflectance. We followed van der Tol et al. (2016) to utilize the
334 Matlab function "lsqnonlin" to minimize the difference between simulated and observed airborne
335 hyperspectral reflectance through optimization interactions. In this process, we obtained the optimized
336 model parameters (crop traits) when the improvement of the cost function (the square sum of the

337 difference between the simulated and observed spectral reflectance) was less than 10^{-3} (Yang et al., 2019).
338 For sun-target-observer angles, we simplified model inversion and set the same observer viewing angles
339 for all airborne imagery pixels to be the nadir, as two HySpex sensors' fields of view are small (17°). We
340 calculated the airborne flights' solar angles according to the site GPS location and the acquisition times.
341 Regarding the direct and diffuse incoming solar radiation spectra, we used MODTRAN (Berk et al.,
342 2005) with the atmospheric visibility and water vapor parameters retrieved from airborne hyperspectral
343 data (Verhoef et al., 2018) for spectral simulations used as SCOPE inputs. The SCOPE model can
344 explicitly quantify plant biomass using the product of LAI and C_{dm}, but not plant nitrogen traits. As plant
345 chlorophyll content is highly linked with nitrogen content (Wang et al., 2021a; Bojović and Marković,
346 2009), we used the RTM-retrieved C_{ab} to explore the relationship between the RTM-based canopy
347 chlorophyll content ($LAI \times C_{ab}$) and the measured cover crop canopy nitrogen content (BMn).
348
349 **2.4.3 Process-guided machine learning**
350 To leverage the domain knowledge from RTMs, the PGML model was pre-trained using RTM-based
351 synthetic simulations with various vegetation and soil conditions. The large-volume synthetic data from
352 RTM simulations enabled the pre-trained PGML model (RTMpt) to be robust and suitable for various
353 environmental conditions. Here, we used long short-term memory networks (LSTM) to develop the pre-
354 train models, as it shows high accuracy to exploit hyperspectral reflectance for predicting agroecosystem
355 variables (Wang et al., 2022; Riese and Keller, 2020). Fig 3(c) shows the LSTM architecture, which used
356 356 airborne hyperspectral reflectance data as inputs and set the spectral bands as LSTM time stamps.
357 Further, we also added a fully connected layer with 64 hidden dimensions to predict cover crop variables.
358 Meanwhile, to improve the model performance for the localized conditions, field measurements of cover
359 crop traits were used for model fine-tuning. Similar to PLSR, we conducted the leave-one-field-out cross-
360 validation to test PGML performance (Fig. 3a). Besides that, we also explored the different fine-tuning
361 data sizes from 10% to 90% (increase 10% each time) to evaluate the impacts of fine-tuning data sets on
362 model prediction performance (Fig. 3b). These experiments conducted for PLSR and PGML can provide

insights into understanding the difference between PLSR and PGML. Furthermore, unlike the data-driven approaches fitting for different target variables, RTMs have a limited number of predicted variables (Table 3). To test the prediction of cover crop variables beyond RTM simulations, we also fine-tuned the pre-trained PGML model (e.g., canopy chlorophyll content) to predict other relevant variables (e.g., canopy nitrogen content). Specifically, as plant chlorophyll content is closely related to nitrogen content, we used the pre-trained model on canopy chlorophyll content ($C_{ab} \times LAI$) to adapt it to quantify BMn in the PGML experiment.

In the model implementation, the PGML model was developed in PyTorch Lightning, a modularized extension of PyTorch. We first split the RTM synthetic dataset with 500,000 records (RTM parameters and their ranges shown in Table 3) into training, validation, and testing with split ratios of 60%, 20%, and 20%. We then trained a one-layer LSTM model followed by a fully connected layer on the training dataset (Fig. 3c) and recorded the R^2 and RMSE values on the validation set after every epoch of the training run. To avoid negative predictions of cover crop varaibles, we modified the cost function to incorporate the non-negativity soft constraint term. After pretraining, we finetuned the LSTM block and the fully connected layers on airborne hyperspectral reflectance and traits. In the fine-tuning process, we performed the train-test split procedure 20 times with different random seeds to finetune and validate our model. In all LSTM training runs, the learning rate is determined by the automatic algorithm (Smith, 2015). During pre-training stage, we used a batch size of 1024 and stop training when the validation R^2 value does not have an increase of at least 0.005 in 20 epochs. During the finetuning stage, we used a batch size of 16 and finetune the network for 100 epochs. Further, we analyzed the permutation-based feature importance to interpret the PGML deep learning model (Wang et al, 2022). Specifically, we randomly permuted each wavelength in the input dataset, fed each permuted dataset to the model, and measured the performance drop with respect to the model performance on the original dataset. As such, we can evaluate the linkage between spectral signatures and predicted variables.

389 **2.5 Model evaluation and error analysis**

390 To comprehensively compare and evaluate these four approaches, i.e., PLSR, RTMn, RTMpt, and
391 PGML, multiple statistics including the coefficient of determination (R^2), Root-Mean-Square Errors
392 (RMSE), relative RMSE (RMSE/range), bias, relative bias (bias/range) were used to evaluate model
393 performance on predicting cover crop traits. PLSR has the flexibility to fit hyperspectral reflectance with
394 various cover crop traits. Thus, we first explored the accuracy of PLSR models on predicting all measured
395 cover crop traits, i.e., BM, N_{mass}, BMn, C_{mass}, BMc, and C/N. Furthermore, As RTM can only predict a
396 limited number of variables (Table 3), we focused on the comparison of PLSR, RTM, and PGML in
397 predicting BM and BMn. Table 4 shows the comparison scheme of these four approaches PLSR, RTMn,
398 RTMpt, and PGML. PLSR used the target cover crop traits for model training and can directly predict
399 BM and BMn. However, RTMn and RTMpt in this study can directly predict only BM using the product
400 of LAI and C_{dm}, but not BMn. To predict BMn, we used the RTM predicted C_{ab} to infer its ability on
401 quantifying cover crop nitrogen traits, as C_{ab} and nitrogen traits are highly correlated for plants (Wang et
402 al., 2021a; Bojović and Marković, 2009). Specifically, we used RTM simulated canopy chlorophyll
403 content to approximate canopy nitrogen content though linear regression with leave-one-field-out cross-
404 validation. For PGML, we used RTM simulated synthetic datasets on BM (LAI × C_{dm}) and canopy
405 chlorophyll content (LAI × C_{ab}) to develop the pre-trained models (RTMpt). Furthermore, the ground
406 truth of BM and BMn were used for fine-tuning pre-trained BM and plant chlorophyll content models,
407 respectively.

408 To compare performance of PLSR, RTMn, RTMpt, and PGML, we used Taylor Diagram (Taylor, 2001)
409 including multiple statistics such as correlation coefficient, normalized standard deviations, and unbiased
410 root mean square errors to visualize the model performance. The diagram shows the correlation
411 coefficient, normalized standard deviation, and normalized unbiased RMSE in angular, radial, and
412 concentric semi-circle coordinates, respectively. The measurement point represents a correlation

coefficient equal to 1, normalized standard deviation equal to 1, and normalized unbiased RMSE equal to 0. Model predictive performance is better when the points are closer to the measurement point.

Furthermore, we analyzed the relationships between their prediction errors and environmental factors. Specifically, we used the boxplots to check the model bias (prediction minus observation) with data collection dates, field crop residue cover, field soil properties (Clay percentage, Silt percentage, and soil organic matter concentration) and measured biomass. Data collection dates were categorized into March and April. Field crop residue cover was approximated by using Cellulose Absorption Index (CAI, Nagler et al., 2003), which is based on the theory that crop residues show different absorption signatures with soil and green vegetation at 2100 nm (Daughtry et al., 2005; Serbin et al., 2009). We obtained the soil properties including clay and silt fractions, and soil organic matter concentration from Gridded Soil Survey Geographic (gSSURGO) Database (Soil Survey Staff, 2018).

To compare the influences of measured data sizes on model performance, we conducted an experiment (Fig. 3b) using various data sizes (from 10% to 90%, each time increasing 10%) for PLSR model training or PGML model fine-tuning. To reduce uncertainties caused by randomness in data splitting, we conducted 20 iterations of data splitting for this experiment.

Table 4. Comparison of four approaches i.e., partial least-squares regression (PLSR), numerical optimization from soil-vegetation radiative transfer modeling (RTMn), pretrained long short-term memory network with RTM synthetic data (RTMpt), and fine-tuned process-guided machine learning (PGML) for quantifying cover crop traits. *RTM predicts chlorophyll content, not nitrogen content. Thus, we used RTM simulated canopy chlorophyll content to approximate canopy nitrogen content though linear regression with leave-one-field-out cross-validation.

Cover crop traits	Unit	PLSR	RTMn	RTMpt	PGML
Aboveground dry biomass (BM)	$g \cdot m^{-2}$	BM	$C_{dm} \times LAI$	$C_{dm} \times LAI$	$C_{dm} \times LAI$
Canopy nitrogen content (BMn)*	$g \cdot m^{-2}$	BMn	$C_{ab} \times LAI$	$C_{ab} \times LAI$	$C_{ab} \times LAI$

435

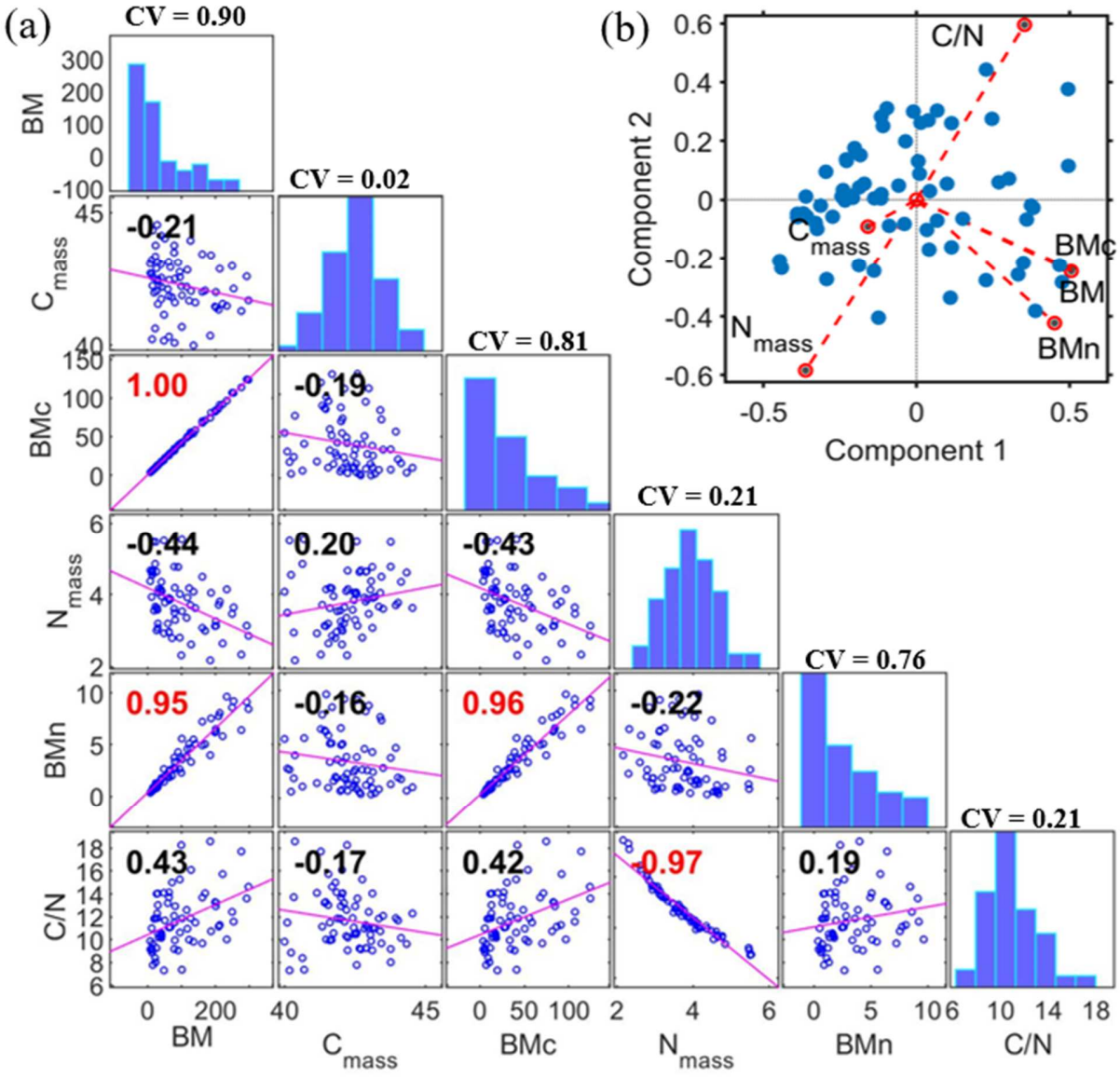
436

437 **3. Results**

438 **3.1 Measured cover crop traits**

439 The correlation matrix among measured cover crop traits (Fig. 4a) shows high correlations for four pairs
440 including BM and BMc, BM and BMn, BMc and BMn, and C/N and N_{mass} (R^2 highlighted with the red
441 color in Fig. 4a). Such correlations indicate that cover crop BMc and N are largely influenced by BM
442 instead of C_{mass} and N_{mass}, respectively. Cover crop plant tissue C/N is mainly influenced by plant N_{mass},
443 not C_{mass}. N_{mass} has a weak negative correlation with BM. From the coefficient of variation (CV, standard
444 deviations/mean), we can see that BM has the highest variability and CVs of BMc and BMn are closer to
445 the CV of BM rather than those of C_{mass} and N_{mass}. The CV of N_{mass} is higher than that of C_{mass}, which is a
446 static variable and has the lowest variability among all variables. These analyses imply that BM is the
447 most important variable to determine cover crop carbon and nutrient dynamics, as supported by the PCA
448 analysis (Fig. 4b). From the PCA component 1 and 2, BM, BMc, and BMn have very similar values.

449



450

451 **Fig 4.** (a) Correlation matrix for measured cover crop traits including cover crop aboveground biomass

452 (BM, g·m^{-2}), plant carbon concentration (C_{mass}, %), plant carbon content per ground area (BMc, g·m^{-2}),

453 plant nitrogen concentration (N_{mass}, %), plant nitrogen content per ground area (BMn, g·m^{-2}), and plant

454 tissue carbon to nitrogen ratio (C/N). The high correlation coefficients are highlighted with the red color.

455 The coefficient of variation (CV) refers to the standard deviation of samples divided by the sample mean.

456 The higher CV values indicate high variabilities of analyzed cover crop traits, or vice versa. (b) Principal

component analysis (PCA) for all cover crop traits. The diagram shows the first two components from the PCA analysis. Similar cover crop traits have close PCA components 1 and 2 values.

3.2 Cover crop traits quantified by Partial Least-Squares Regression

With leave-one-field-out cross-validation, PLSR shows different performance to quantify cover crop traits, i.e., BM, C_{mass}, N_{mass}, C/N, BMc, and BMn (Fig. 5), from airborne hyperspectral reflectance. Among six cover crop traits, PLSR achieved the highest model performance on predicting BM (R^2= 0.72, relative RMSE = 17.23.%). As highly correlated with BM (Fig. 3), PLSR also obtained high prediction performance on BMc (R^2= 0.71, relative RMSE = 17.70%) as well as BMn (R^2= 0.60, relative RMSE = 21.52%). Airborne hyperspectral PLSR models can quantify BM and BMc well, but not C_{mass}. This finding confirms with Fig. 4 analysis, which shows that C_{mass} is a static variable and BM determines the variability of BMc. Airborne hyperspectral reflectance signals are insensitive to quantify cover crop C_{mass} (R^2= 0.02, relative RMSE = 29.13%). However, this does not influence the accurate quantification of BMc from remote sensing. Meanwhile, BMn is influenced by both BM and N_{mass}. The prediction of BMn achieved slightly weaker performance than for BMc could be due to the weaker model prediction of N_{mass} (R^2= 0.31, relative RMSE = 23.73%). Similarly, as C_{mass} is more static, the C/N model performance (R^2= 0.25, relative RMSE = 21.89%) is closer to the model performance of N_{mass}. Overall, hyperspectral signals have the capability to detect biomass and nitrogen concentration, but are insensitive to detect cover crop carbon concentration. Also given the predictable variables from RTM, we tested RTMn, RTMpt, and PGML retrieval on BM and BMn.

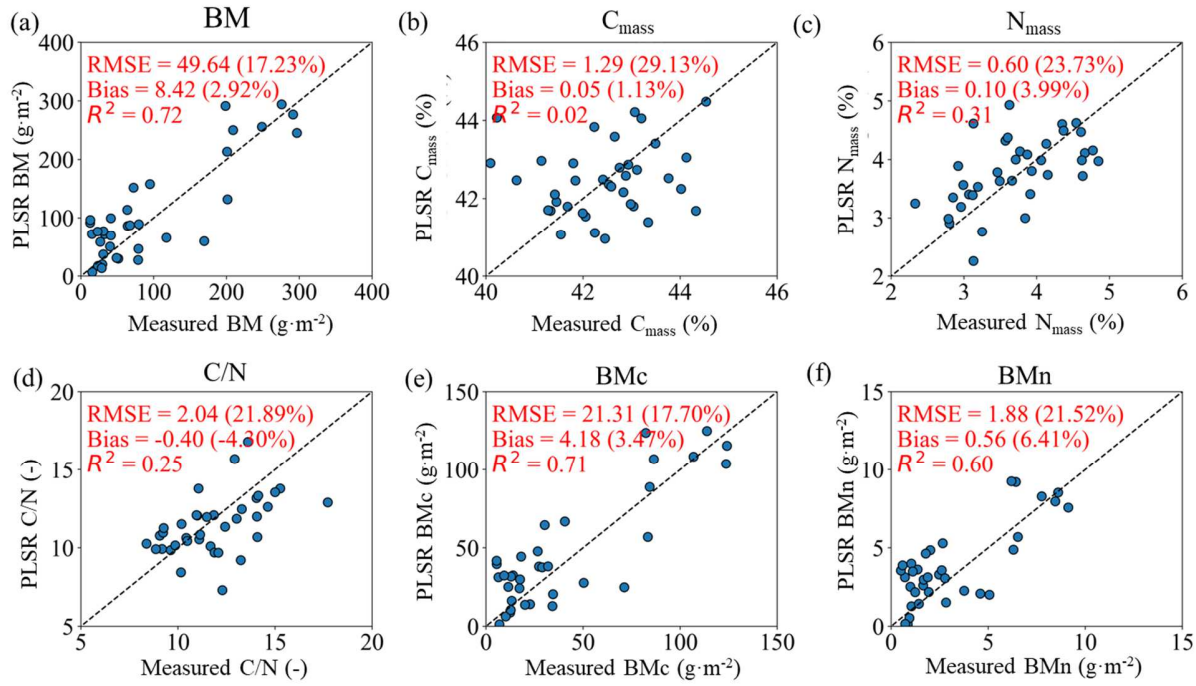


Fig. 5 Partial Least-Squares Regression (PLSR) based leave-one-field-out cross-validation of predicting cover crop aboveground (a) biomass (BM, g·m^{-2}), (b) carbon concentration (C_{mass}, %), (c) nitrogen concentration (N_{mass}, %), (d) carbon to nitrogen ratio (C/N), (e) canopy carbon content (BMc, g·m^{-2}), and (f) canopy nitrogen content (BMn, g·m^{-2}) from airborne hyperspectral imagery. The leave-one-field-out cross-validation refers to that in cross-validation, data from the same field and date were used in model either training or testing, but not both. As such, we can avoid that the similar data from the same field and date were used for both training and testing.

As a typical field shown in Fig. 6, we applied the PLSR models to quantify the spatial variabilities of cover crop traits (the mean values of predictions from all leave-one-field-out cross-validation models) as well as their uncertainties (the standard deviations of predictions from all cross-validation models). Overall, the quantified cover crop traits show high spatial variations within the fields. This indicates that cover crop growth in the study region has significant variability and high-resolution remote sensing data are indeed needed to capture such variability. The biomass-related traits including BM (Fig. 6b), C/N (Fig. 6d), BMc (Figure 6e), and BMn (Fig. 6f) have very similar spatial patterns, as biomass controls

these variables. Meanwhile, N_{mass} has different spatial patterns, and areas with higher biomass tend to have lower N_{mass}. These quantified cover crop traits can be valuable to inform stakeholders to take decisions on cover crop management, e.g., termination time.

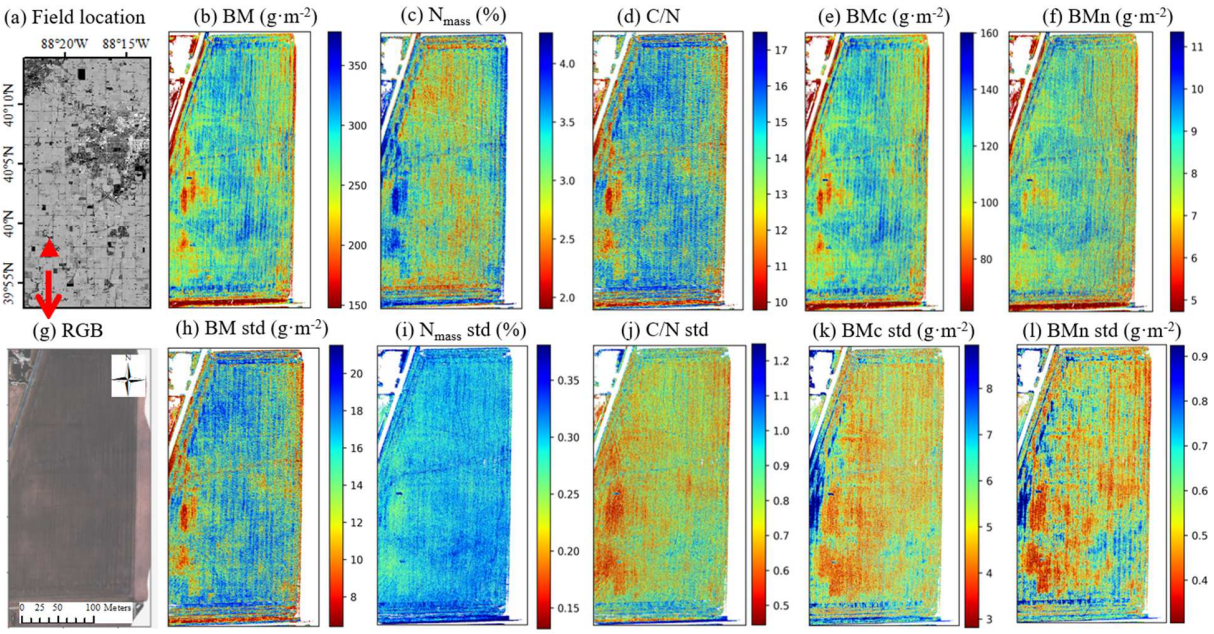


Fig. 6 Airborne hyperspectral imagery(a, g) based PLSR predictions of cover crop (b) aboveground biomass (BM, $g \cdot m^{-2}$), (c) nitrogen concentration (N_{mass}, %), (d) carbon to nitrogen ratio (C/N), (e) carbon content (BMc, $g \cdot m^{-2}$), (f) nitrogen content (BMn, $g \cdot m^{-2}$), and uncertainties for (h) biomass (BM std, $g \cdot m^{-2}$), (i) nitrogen concentration (N_{mass} std, %), (j) carbon to nitrogen ratio (C/N std), (k) carbon content (BMc std, $g \cdot m^{-2}$), (l) nitrogen content (BMn std, $g \cdot m^{-2}$) on April 13th, 2021. Soil pixels with NDVI < 0.35 were masked out (white color).

3.3 Cover crop traits quantified by radiative transfer modeling and process-guided machine learning

We quantified cover crop BM and BMn through RTMn, RTMpt, and PGML (Fig. 7). Among these approaches, RTMn and RTMpt show similar model performance in predicting cover crop biomass. Meanwhile, PGML shows better performance ($R^2 = 0.72$, relative RMSE = 15.16%) in leave-one-field-out

cross-validation as incorporating field data for model fine-tuning. PGML can mitigate the imperfection in RTM e.g., lacking clumping correction for the 4SAIL model (the SCOPE submodel of simulating canopy reflectance). 4SAIL assumes uniform distribution inside the plant canopy, however, grassland leaves often are highly clumped (Chen et al., 2005). Furthermore, PGML (Fig. 7e) also shows slightly higher performance than PLSR (Fig. 5a, R^2= 0.72, relative RMSE = 17.23%) in predicting BM. Similarly, cover crop BMn were better quantified through PGML (Fig. 7f, R^2= 0.69, relative RMSE = 16.59%) than PLSR (Fig. 5f, R^2= 0.60, relative RMSE = 21.52%).

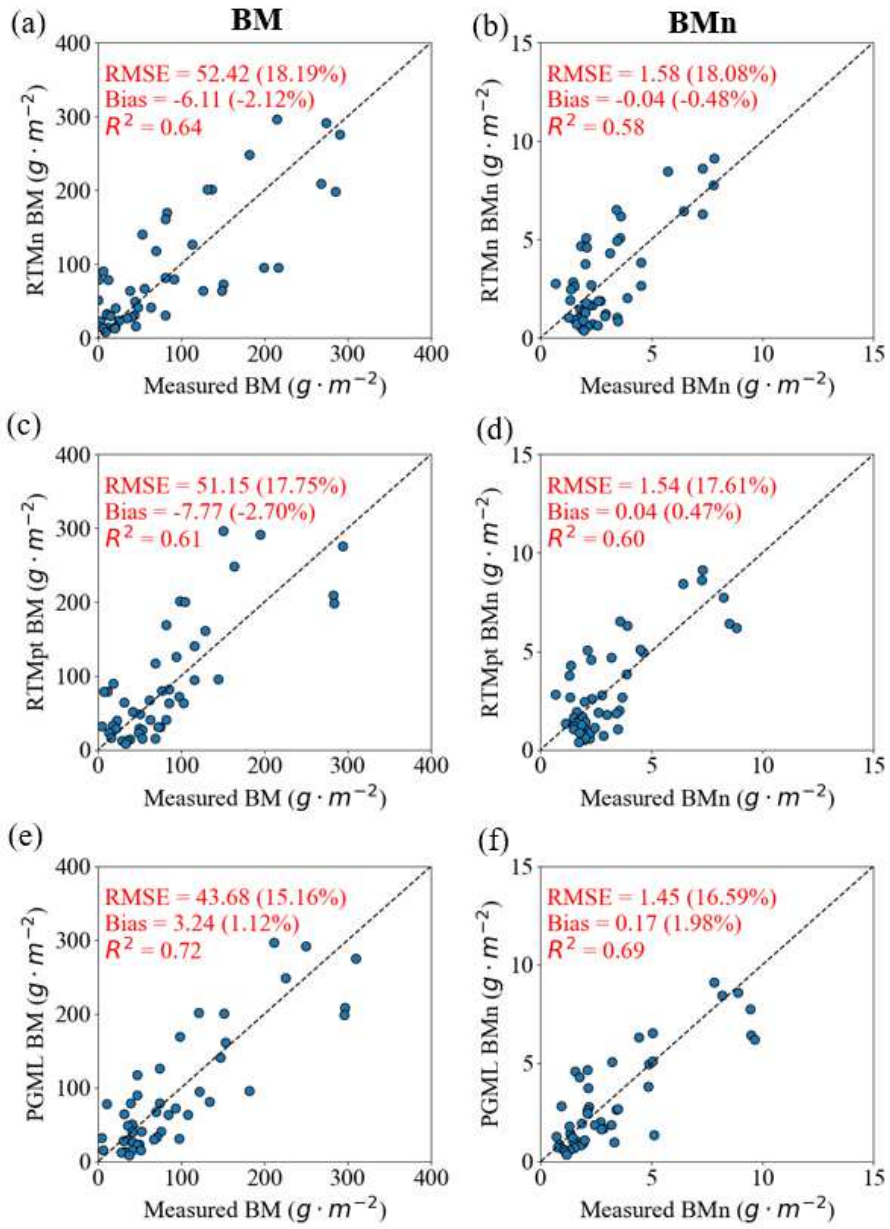
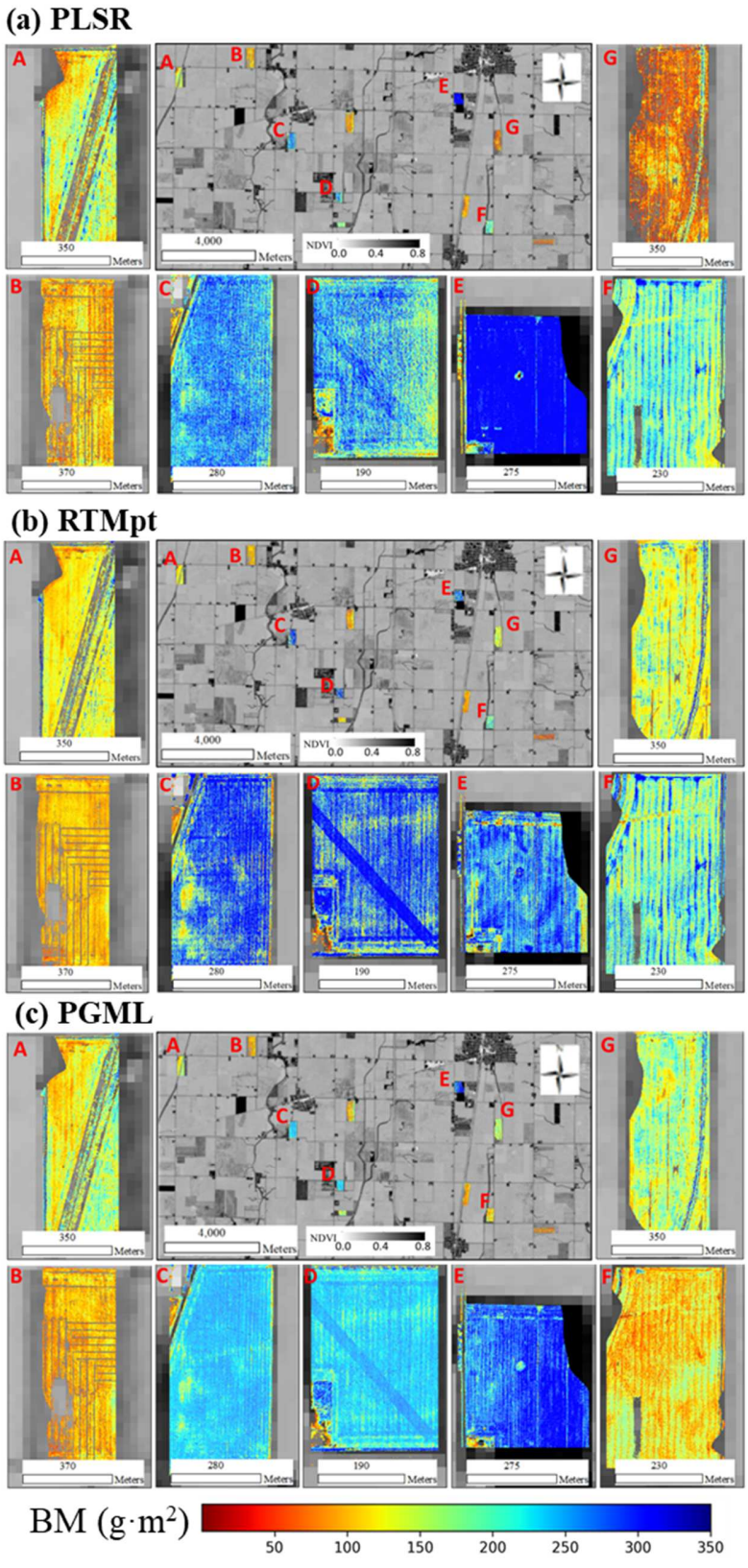


Fig. 7 Retrieval of cover crop (a, c, e) aboveground biomass (BM), and (b, d, e) canopy nitrogen content (BMn) from airborne hyperspectral reflectance by (a, b) radiative transfer model numerical inversion (RTMn) and (c, d) RTM surrogate model by long short-term memory networks, (e, f) process-guided machine learning (PGML) with leave-one-field-out cross-validation.

Through applying PLSR, RTMpt, and PGML to airborne hyperspectral imagery, we obtained BM estimates for all survey fields. We did not apply RTMn to the whole airborne hyperspectral imagery to quantify spatial variability of cover crop traits, due to its large computation needs for numerical optimization. Fig. 8 shows the BM for selected typical fields in the study region. The quantified BM from the three approaches have similar spatial patterns within each field, however, their BM values in a few fields could be different. We found that the quantified BM values from PGML are often between PLSR and RTMpt estimates, for example, fields A, B, C, D, E, and G in Fig. 8, as PGML integrates both RTM simulations and field measurements. Furthermore, in a few fields with low biomass (e.g., field G in Fig. 8), PLSR tends to have lower estimates than RTMpt and PGML. Meanwhile, in fields with high biomass (e.g., field E in Fig. 8), PLSR's estimates are higher than those from RTMpt and PGML. Within a field (e.g., field A in Fig. 8), PLSR also shows lower values in the northwest low BM regions of the field, but higher values in middle parts of the field than RTMpt and PGML. The higher variability from PLSR but lower variability from RTMpt and PGML could be due to the fact that RTMpt and PGML were constrained by a large number of synthetic data simulations from RTM, while PLSR was trained by limited field measurements. Overall, from all three approaches, we found that there were significantly large variabilities of BM with values ranging from 0 to 350 $g \cdot m^{-2}$ within and across fields. Such variability could result from the different management practices (planting dates, seeding density) and soil conditions (soil nutrient and water availability) in the fields. This also highlights the importance of utilizing high-resolution hyperspectral imagery rather than coarse-resolution remote sensing data to accurately quantify cover crop biomass.


(a) PLSR
(b) RTMpt
(c) PGML
A
B
C
D
E
F
G
4,000
Meters
NDVI
0.0
0.4
0.8
350
370
280
190
275
230
BM (g·m²)
50
100
150
200
250
300
350

543

Fig. 8 Comparison of quantified cover crop aboveground biomass (BM, $g \cdot m^{-2}$) from (a) PLSR, (b) RTMpt, (c) PGML in a few selected fields (A-H) in the study region on April 13th, 2021. The base map is satellite 30-m NDVI on April 13th, 2021 from the STAIR fusion algorithm (Luo et al., 2018) and the dark color refers to a higher NDVI value. The red polygon refers to the surveyed cover crop fields in this study. In the biomass maps, soil pixels with NDVI < 0.35 were masked out.

3.4 Model comparison and error analysis

To comprehensively evaluate PLSR, RTMn, RTMpt, and PGML, we used multiple statistics in Taylor Diagram (Fig. 9) to compare their predictions on cover crop biomass and nitrogen content. Due to RTMpt and RTMn are the direct inversion of RTM simulations without incorporating field measurements, both approaches have very similar model performance regarding correlation coefficient, standard deviations, and unbiased RMSE. PLSR results for biomass and nitrogen content have similar performance. Among all, PGML shows a closer distance to the measurements and outperformed all other approaches to predict both biomass and nitrogen content.

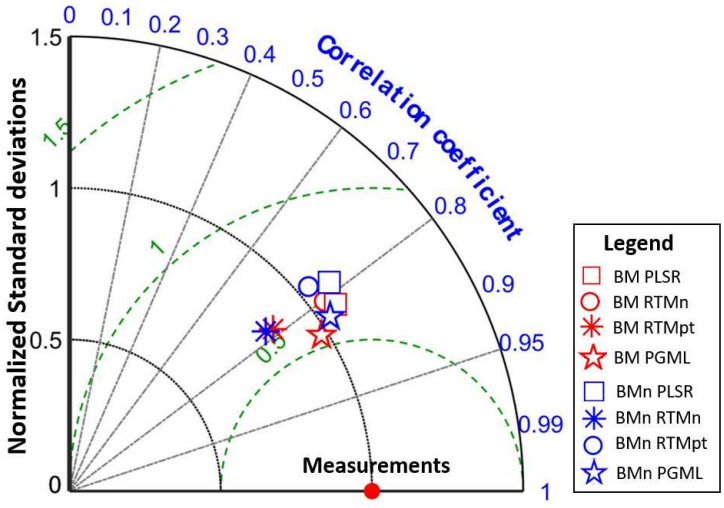
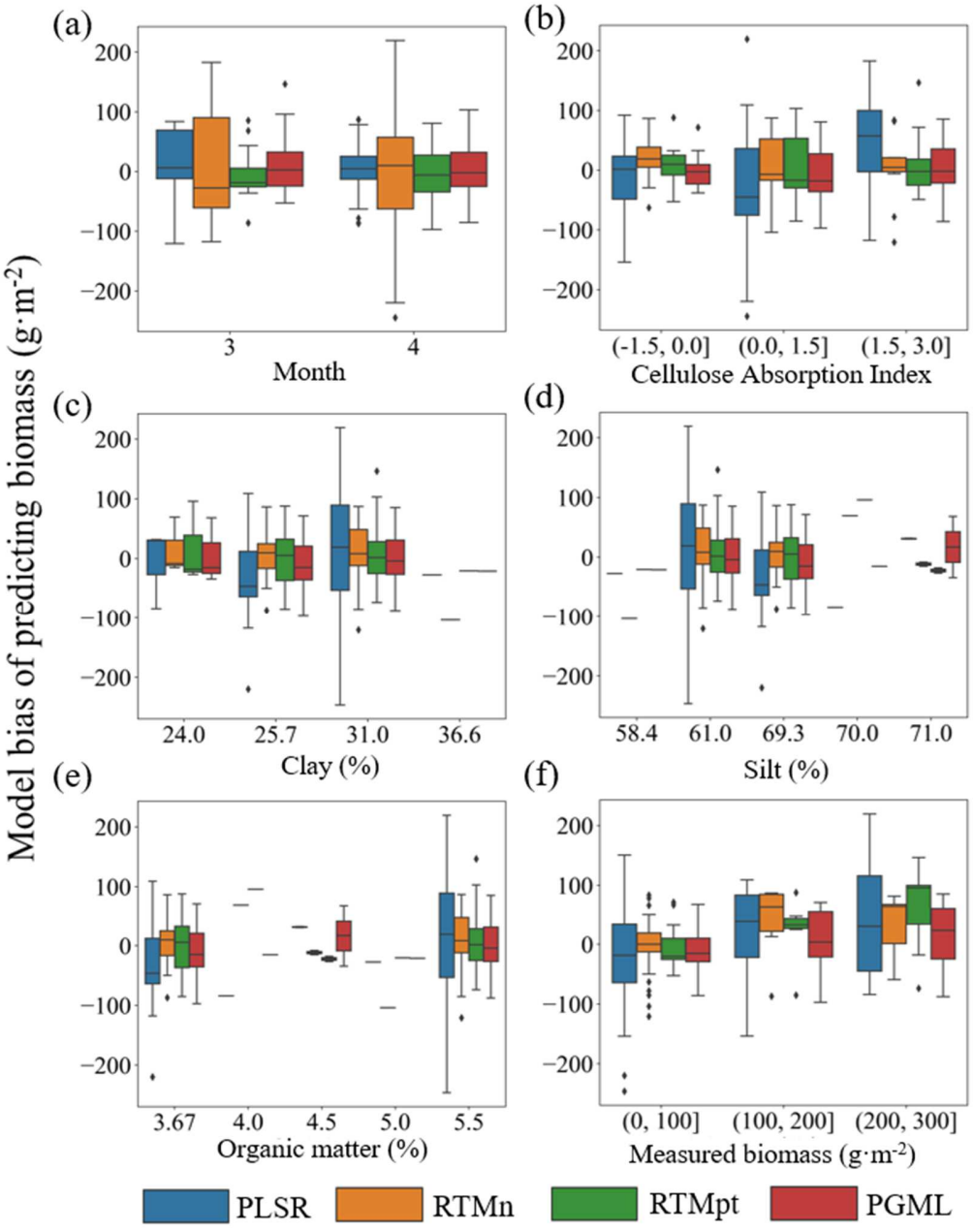


Fig. 9 Taylor Diagram to compare the model predictions of cover crop aboveground biomass and canopy nitrogen content from partial least-squares regression (PLSR, the square sign), numerical optimization from radiative transfer modeling (RTMn, the star sign), pretrained machine learning model from radiative

562 transfer modeling (RTMpt, the circle sign), and process-guided machine learning from radiative transfer
563 modeling (PGML, pentagram sign). The diagram shows the correlation coefficient, normalized standard
564 deviation, and normalized unbiased RMSE in angular, radial, and concentric semi-circle coordinates,
565 respectively. The measurement point represents a correlation coefficient equal to 1, normalized standard
566 deviation equal to 1, and normalized unbiased RMSE equal to 0. Model predictive performance is better
567 when the points are closer to the measurement point.
568
569 To understand the differences among PLSR, RTMn, RTMpt, and PGML, we analyzed the model errors
570 with environmental factors (Fig. 10). In comparing sampling dates (Fig. 10a), we found that RTMn shows
571 the large differences (higher value ranges in boxplots) between March and April predictions. In different
572 crop residue cover (Fig. 10b), soil texture (Fig. 10c, d), soil organic matter (Fig. 10e), and measured cover
573 crop biomass (Fig. 10f), PLSR shows larger variabilities in model errors than other approaches.
574 Meanwhile, RTMn, RTMpt, and PGML approaches have more consistent model errors across different
575 categories of environmental factors. Overall, among the four approaches, we found that PGML has the
576 lowest and consistent errors across fields with different conditions.
577



578

579 **Fig. 10**. Boxplots for analyzing the relationship between cover crop biomass model biases (prediction
580 minus observation) with different variables including (a) collection date (b) Cellulose Absorption Index
581 (CAI, a proxy of crop residue cover. The higher CAI value refers to a higher fraction of crop residue
582 cover), (c) soil clay fraction, (d) soil silt fraction, (e) soil organic matter concentration, and (f) measured
583 cover crop aboveground biomass.

584

3.5 Model sensitivity to training data sizes

In testing the sensitivity of model performance to the training data sizes (Fig. 11), we found that the model performances of PGML were consistently higher than those of PLSR for quantifying BM and BMn with different sizes of fine-tuning or training datasets. The model performance for predicting cover crop BM and BMn significantly improved with the increased percentages of field data in both PGML and PLSR. Meanwhile, we also observed that after adding 30% of field data for BM, PGML performance did not gain improvement as much as the model performance changes from 0-30%. Such findings indicate that after a certain threshold, the PGML model turns to be consistent and the benefits of adding field measurements to PGML become marginal. Similarly, the threshold for PLSR BM was also observed with a value of approximately 50%, which is higher than PGML and indicates that PGML requires less data for model tuning than PLSR.

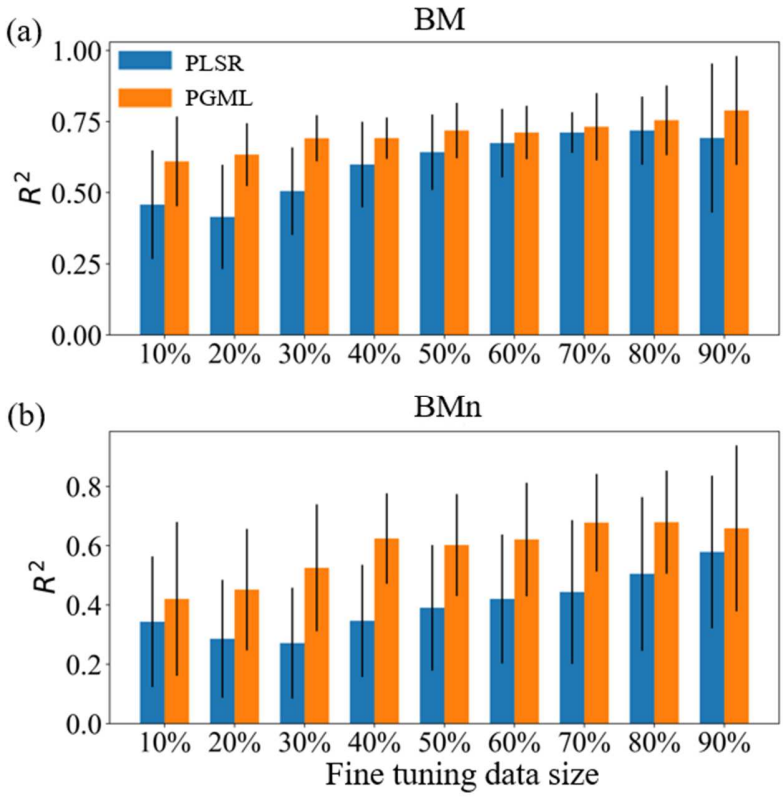


Fig. 11. Model performance of quantifying cover crop (a) aboveground biomass (BM) and (b) canopy nitrogen content (BMn) with different training or fine-tuning data sizes. Blue bars refer to PLSR model

599 performance and orange bars are for PGML. The error bars refer to the standard deviations resulting from
600 20 times random data splitting.
601
602 **3.6 Spectral linkage among cover crop traits**
603 To understand the spectral linkage among cover crop traits, we analyzed the feature importance VIP
604 scores of PLSR models as well as RTMpt and PGML permutation feature importance (Fig. 12). In the
605 VIP scores of PLSR models, variables including BM, BMc, and BMn had similar feature importance with
606 the highest feature contribution from the blue, green, red edge, and shortwave infrared wavelengths. The
607 similar feature importance among BM, BMc, and BMn also demonstrate that BM drives the variability in
608 BMc and BMn (Fig. 3). Meanwhile, N_{mass} and C/N were more similar with the most important feature in
609 the red edge around 780 nm, green spectra, and shortwave infrared. By comparing RTMpt and PGML
610 results, we found that fine-tuning with field data can significantly alter model feature importance. Before
611 fine-tuning, RTMpt show higher model feature importance in the shortwave infrared range, as BM in
612 RTMpt was calculated by the product of LAI and Cdm. However, after fine-tuning, PGML has higher
613 contributions from red edge and green contributions for predicting biomass, which indiates that red edge
614 and green spectra are helpful to distinguish cover crop biomass with soil or crop residue background
615 signals. We also found PGML-based feature importance values were overall similar to the VIP scores
616 from PLSR, but they also showed certain differences. Both feature importance had high values in the red
617 edge around 710 nm and shortwave infrared around 2000 nm. However, PGML had higher relative
618 importance features in green spectra (around 540 nm) rather than blue spectra in the PLSR VIP scores. In
619 view of previous studies only using NDVI to quantify cover crop biomass (e.g., Muñoz et al., 2010;
620 Breunig et al., 2020), this spectral feature analysis indicates the model accuracies of predicting cover crop
621 traits could be further strengthened through adding red edge reflectance, for example, by using data from
622 missions like Sentinel-2.
623

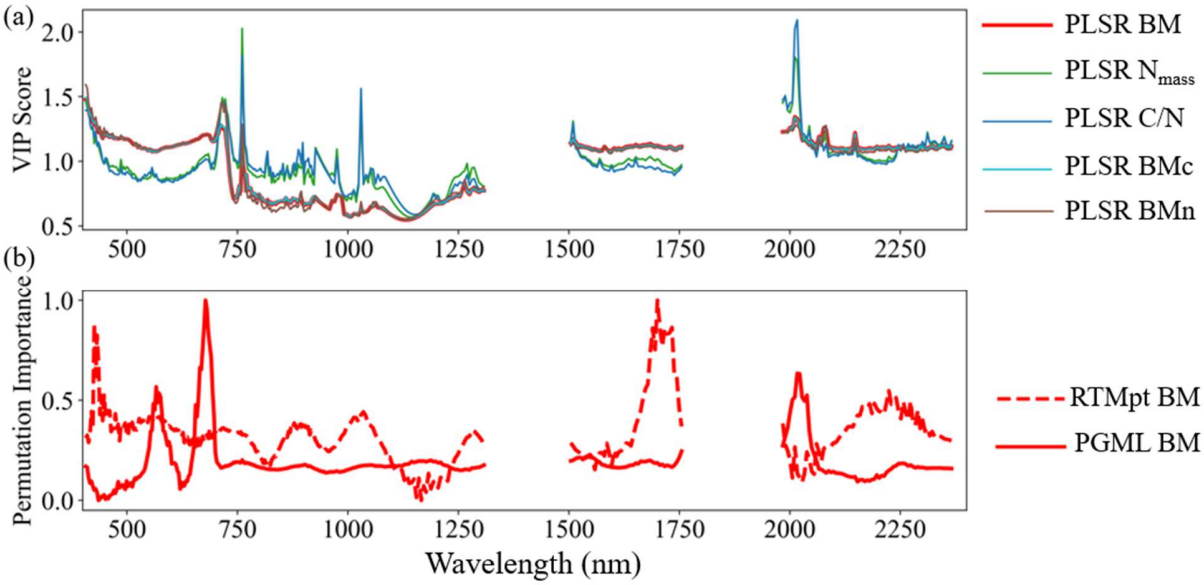


Fig. 12 Feature importance for (a) Partial least-squares regression (PLSR), and (b) the pretrained model from radiative transfer modeling (RTMpt) and process-guided machine learning (PGML) fine tuned with field measurements. (a) The y-axis refers to the Variable Influence on Projection (VIP) scores of hyperspectral PLSR models for predicting cover crop aboveground biomass (BM), carbon concentration (C_{mass}), nitrogen concentration (N_{mass}), carbon and nitrogen ratio (C/N), carbon content (BMc), and nitrogen content (BMn). (b) The y-axis is the permutation importance for RTMpt and PGML. The higher values of VIP scores and permutation importance indicate higher feature importance.

4. Discussion

This study provided the first exploration of utilizing airborne hyperspectral imaging to quantify the growth of cover crops, an important conservation practice for sustainable agriculture in regions like the U.S. Corn Belt. Moreover, we demonstrated that an emerging modeling paradigm PGML, which integrates the strengths of data-driven machine learning and process-based radiative transfer modeling, has a high capability for remote sensing-based retrieval of plant traits. Among cover crop traits, we found that airborne hyperspectral data with PGML most accurately quantified cover crop biomass and nitrogen content. In further discussion, we summarized the benefits of airborne hyperspectral imaging techniques

for cover crop monitoring, and analyzed the utility of PGML in improving the existing remote sensing retrieval schemes for quantifying plant traits.

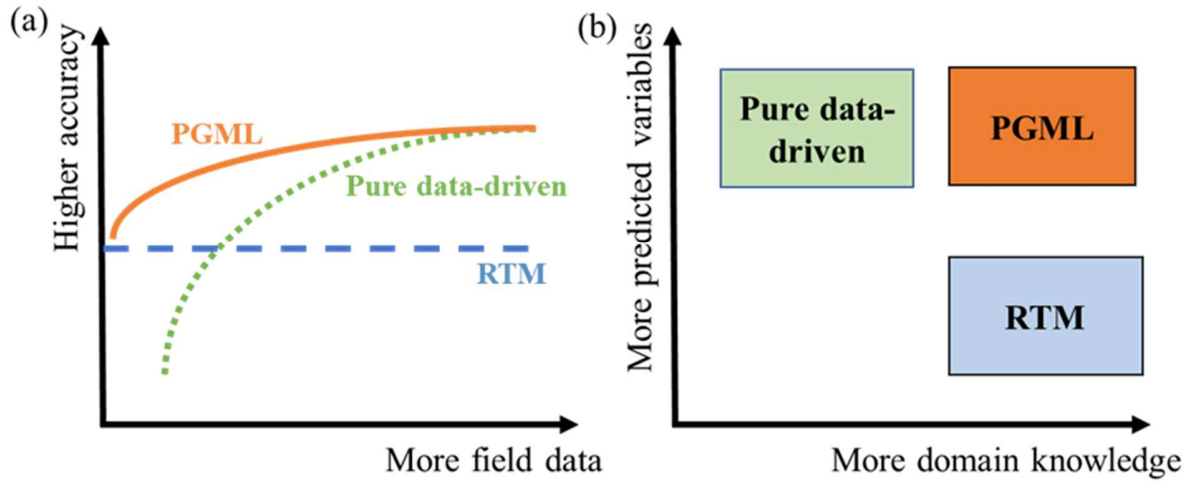
4.1 Airborne hyperspectral imaging of cover crop traits

In this study, we demonstrated the high accuracy of utilizing high-resolution full-optical-range hyperspectral reflectance for quantifying cover crop traits. Quantifying plant traits in partially vegetation conditions with diverse soil background signals is often challenging. Previous multi-site studies (>10 sites) with multispectral data and environment covariates (e.g., air temperature, growing degree days) can achieve limited accuracies, e.g., R^2 around 50–73%, inpredicting cover crop biomass (Goffart et al., 2021; e.g., Hively et al., 2009; Xia et al., 2021). Compared to these studies, this study of using PGML obtained higher accuracies (Fig. 7) of cover crop biomass (relative RMSE = 15.16%) and nitrogen content (relative RMSE = 16.59%) with only remote sensing data as inputs. There are three major reasons for achieving such higher accuracies. The airborne hyperspectral imagery contains rich spectral information with 356 wavelengths, which is associated with diverse plant traits. Furthermore, given the large spatial variabilities of cover crop traits within and across fields (Fig. 6 and 8), the 0.5m high-resolution imagery pixels can be utilized to seamlessly match with field data from small sampling plots. In addition, this study also leveraged advanced machine learning and soil-vegetation radiative transfer modeling to exploit airborne hyperspectral data for quantifying cover crop traits. Particularly, soil-vegetation radiative transfer modeling can separate soil and vegetation signals to facilitate the quantification of vegetation traits from mixed remote sensing data. Besides these advantages, airborne hyperspectral techniques also have limitations on spatial and temporal coverage, compared to high-resolution satellite data (e.g., Landsat and MODIS fused STAIR satellite data, Luo et al., 2018). To provide large-scale timely information on cover crop growth, cross-scaling sensing to integrate airborne hyperspectral remote sensing and satellite data has high potential. Specifically, the airborne hyperspectral imaging system can be used as a “ground truth” collector to scan cover crop fields to obtain highly accurate and large datasets of cover crop traits. Furthermore, these airborne-based quasi “ground truth” data will be integrated with

multi-source satellite fusion data, e.g., STAIR, to obtain daily and 10-30 m estimates of cover crop traits. As such, we can cost-effectively and accurately quantify cover crop growth in every field across the region of interest. With the large-scale cover crop growth information, we can provide timely information to guide growers' management practices e.g. termination as well as support policymakers to verify cover crop growth to assess the outcomes of cover crop conservation programs.

4.2 Radiative transfer process-guided machine learning

Traditionally, the remote sensing community uses either process-based soil-vegetation radiative transfer modeling or data-driven machine learning approaches to quantify plant traits from surface reflectance. Process-based models are built based on radiative transfer differential equations on simulating photon reflectance, transmission, and scattering for each canopy or soil layer. Meanwhile, machine learning used data mining and fitting approaches to identify relationships between input features and target variables. As illustrated in Fig. 13, by leveraging the strengths of both process-based models and data-driven approaches, PGML can efficiently reduce data needs for machine learning training, incorporate domain knowledge on radiative transfer modeling, increase the model transferability for various environmental conditions, and potentially achieve higher model accuracy than process models and pure data-driven approaches.



685 **Fig. 13**. Conceptual diagram for comparing process-guided machine learning (PGML), soil-vegetation
686 radiative transfer model inversion (RTM), and pure data-driven machine learning. (a) Comparison for
687 field data demand and model predictive performance. (b) Comparison for domain knowledge
688 incorporation and the number of variables in model predictions.
689
690 Compared to RTMpt and RTMn, PGML has the benefits of achieving high predictive model performance,
691 reducing uncertainties from RTM imperfection, and mitigating spectral input noises. Specifically, due to
692 incomplete knowledge of certain processes and model simplification for reducing computation needs,
693 process-based radiative transfer models are approximations of real processes that operate in ecosystems.
694 Significant uncertainties on model retrieval accuracy as well as the limited number of predictable
695 variables constrain the practical applications of radiative transfer models to quantify crop traits from
696 remote sensing. Furthermore, RTM retrieval needs to be operated with highly accurate spectral inputs as
697 the synthetic data sets. However, different sensor types have various sensor signal-to-noise ratios and
698 spectral noises can add challenges to the accurate retrieval of crop traits from spectral data. By
699 incorporating fine-tuning processes into modeling, PGML can potentially overcome these challenges to
700 mitigate RTM imperfection and spectral input noises to achieve high model predictive performance.
701
702 Compared to pure data-driven approaches, PGML can provide benefits in reducing the need for large-
703 volume label data for machine learning and improving machine learning transparency. In past years,
704 machine learning has shown tremendous success in handling large-volume datasets to achieve high
705 accuracies in predicting target variables with input features. However, as a black box, machine learning
706 lacks the ability to interpret the relationships between input drivers and the target variables, and therefore
707 is hard to provide insights into underlying mechanisms. Furthermore, machine learning has risks of
708 emphasizing model fitting and ignoring mechanistic principles. For example, for quantifying cover crop
709 traits from remote sensing data, the mixture of various background signals from soils and crop residues
710 complicated the retrieval of plant traits. The pure data-driven approaches like PLSR may overfit the

711 training data sets and lacks transferability to unmix cover crop and soil background signals across
712 different environmental conditions. Adding the unmixing algorithms from the generalized RTM
713 simulations could enhance the model transferability across environmental conditions. In addition, pure
714 machine learning requires a large volume of data sets for model training. The synthetic data from RTM
715 simulations can provide sufficient data sources for developing the robust pre-train model to reduce the
716 need for field data.
717
718 **5. Conclusion**
719 This study demonstrated the first exploration of utilizing airborne hyperspectral reflectance in the full
720 optical range (400–2400 nm) on quantifying cover crop growth, which is an important conservation
721 farming practice to improve agriculture sustainability for the U.S. Corn Belt. We found the great potential
722 of high-resolution hyperspectral data to accurately quantify cover crop biomass and nitrogen traits, but
723 hyperspectral data are relatively weak in detecting cover crop tissue carbon concentration. The spectral
724 information in red edge, green and shortwave wavelengths are important for quantifying cover crop
725 biomass and biomass nitrogen content. Moreover, this study also demonstrated a new remote sensing
726 retrieval scheme, radiative transfer process-guided machine learning (PGML), to fully capitalize on the
727 rich hyperspectral reflectance information to quantify cover crop traits. PGML utilizes the large-volume
728 synthetic data from soil-vegetation radiative transfer modeling to pre-train the machine learning model.
729 By adding limited field measurements for model fine-tuning, PGML can achieve high accuracy in
730 predicting cover crop aboveground biomass and nitrogen content. PGML integrates the benefits of
731 process-based radiative transfer modeling and data-driven machine learning approaches. Compared to
732 radiative transfer model inversion, PGML can achieve higher predictive performance and can predict
733 other relevant variables, which are not explicit in RTM. Meanwhile, PGML has less dependence on field
734 data for model tuning than partial least-squares regression. Our study highlights the importance of
735 integrating domain knowledge (i.e. soil-vegetation radiative transfer modeling) and machine learning to
736 accurately quantify vegetation variables to guide conservation practices for sustainable agriculture.

737

738

739 **Credit author statement**

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741 L.H.; Data processing and modeling, S.W., C. Z., Q.Z.; Result analysis: S.W., K.G., C. Z., E.A., J.H.,
742 J.W., A.M., ; Writing-Original Draft, S.W.; Writing-Review & Editing, all authors; Visualization, S.W.,
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